Running app: https://documind-web-k6i1.onrender.com

Team 06 — Milestone 5 submission

Field	Value
Running app	https://documind-web-k6i1.onrender.com
Team	Team 06 — DocuMind
Members	To Hien Hai Dang (11247271); Phan Dang Vu (11247373); Bui Dang Duong (11247274); Nguyen Bao Tai (11247348)
Scrum Master	Bui Dang Duong (Sprint 5)
Repository	https://github.com/SE-FDA-NEU/G6_AI66A_SE
Release v1.0	https://github.com/SE-FDA-NEU/G6_AI66A_SE/releases/tag/v1.0
Latest green CI run on main	https://github.com/SE-FDA-NEU/G6_AI66A_SE/actions/runs/43412489569
Test accounts	Standard-user demo account: test@gmail.com / 12345678.

Production operations

1. How to run and release

Public app: https://documind-web-k6i1.onrender.com. Frontend and FastAPI backend run on Render; backend: https://documind-n9ny.onrender.com. MySQL-compatible TiDB and persistent backend files are owned by Duong. Platform configuration still requires owner verification.

Run locally using SETUP. Release: reviewed PR merge successful main CI deploy the same SHA to both services. The frontend blueprint specifies checksPass; Dang must verify the backend gate and Tai the live frontend setting. Keep one backend instance/worker; stop the old process tree before replacement.

Rollback: stop traffic, select the last verified compatible SHA in both Render services, redeploy, then check login, DB, files and citations. Keep the same DB/storage. If schema changes are incompatible, stop and restore into a separate environment before switching; code rollback does not restore data. Record both deploy IDs and smoke results in release acceptance.

2. The CI pipeline

Step	Checks
Setup/install	Python 3.12; Node 24; pip development requirements; npm lockfile; Chromium/WebKit
Python tests	Full pytest; JUnit artifact
Node build and tests	Desktop/Android/iOS Playwright; production configuration, build and session tests; reports
No secrets committed	Credential patterns; no tracked environment files or database dumps
Artifacts	Test/build reports retained 14 days

Latest verified green main: 43412489569, SHA 0431a5d463295f6acd38b87c367e79c47a7037cc. Measured start-to-completion: **6 min 20 s**, 2026-11-13 10:52:34–10:58:54 UTC. Refresh this evidence for the release SHA.

Main protection requires one PR approval and blocks force pushes/deletion. API inspection found no required status checks. An organization administrator must require all three checks and supply a settings screenshot; screenshot and enforcement remain release blockers (#251).

3. Monitoring

Backend /health runs bounded SELECT 1: 200 ready, 503 unavailable; it does not check AI. Verify /health on the public frontend too. Render service Logs contain sanitized request route/status/duration and processing state/error codes. Never log passwords, tokens, keys, personal data, document contents or search text. Use the logging configuration.

Signal	Detection and response	Owner
Health	Check every 60 s; alert after three failures; inspect service/DB	Dang
Error rate	5xx >5% over 5 min, 20 requests; inspect logs/release	Dang
Response time	Ordinary JSON p95 >2 s over 5 min; inspect DB/load	Dang
Jobs/quota/disk	Three job failures/10 min, any quota error, disk >80%; inspect provider/storage	Dang / Duong

Thresholds are proposed. Monitor URL, notification channel and backup responder must be verified before acceptance; no active alert integration is claimed.

4. Known risks and technical debt

Item	Layer	What could happen	Plan
In-memory queue	code	Restart interrupts work; concurrency has no explicit admission cap	#249; single process, failed-job resubmission; capacity verification #248
Persistent files	data	DB survives while originals disappear	#246; mounted storage and coordinated restore rehearsal
AI availability	infrastructure	Quota errors or partial batch failures	#254; retest five-file batch after #269
Required CI checks absent	infrastructure	Broken merge can bypass checks	#251; administrator adds three required checks
Public acceptance incomplete	tests	Local passes hide hosting/browser failures	#257; all nine P0s on public HTTPS
Missing deployment/alert evidence	docs	Operator cannot identify release or detect outage	#256; owners verify service settings and evidence
Cold start	infrastructure	First request is slow	Accepted: grading brief permits slow initial visits; check before demo

5. Maintenance plan

Area	What to do	Owner	How often
Bugs	Triage, reproduce, fix through reviewed green PR	Vu coordinates; component owner fixes	Daily and each report
Dependencies	Review advisories/lockfiles; test before update	Dang backend; Tai frontend	Weekly and urgent advisories
Backups	Quiesce writes/workers/retention; coordinated MySQL + files backup; encrypt off-host copy	Duong	Daily and before release/schema changes
Monitoring	Review health, errors, latency, quotas, disk; respond	Dang; Duong storage	Daily; alerts continuously once configured
Documentation	Recheck URLs, SHAs, commands, accounts, ownership	Vu with Dang/Duong/Tai	Each release and weekly

From backend, use python -m app.services.storage_ops --backup, then --restore-test /secure/bundle. The latter verifies a separate temporary SQLite restore, not MySQL recovery. Rehearse the SQL dump in an empty isolated MySQL database with matching files, then --verify and citation checks before cutover. Never restore over live data blindly. Proposed retention: seven daily encrypted backups; Duong must approve and enforce deletion/access controls.

Documents expire 30 days after successful open/download/update; warn seven days beforehand. Listing/polling does not renew expiry. Cleanup defaults to hourly batches of 100. Backups must follow deletion policy. Queue shutdown defaults to 10 s grace plus 8 s cleanup; host shutdown must allow longer. See restart limits.

6. Lessons learned

Incident	Fix	Future change
Linux CI counted empty .analysis-cache as leftover upload	Assert remaining files recursively, allow empty infrastructure directories (#263)	Include hidden paths in cleanup checks
Mobile CI waited for hidden folder controls	Wait for mobile toggle/sidebar visibility (M4 report)	Test responsive state transitions in CI
Five-file proxy upload exceeded Next's 10 MB default	Set 30 MiB proxy allowance and add 25 MiB production test (#267)	Verify platform limits before deployment
Public batch completed only 3/5 files	#269 changes analysis/key handling; production retest pending	Preserve failed-job evidence and require live AI acceptance (#254)

7. Availability commitment and verification

Proposed accountable coordinator: Vu (@PhanDangVu), until grades are published. The publication date is unspecified; this is an event-based end date. Vu must confirm the commitment, keep hosting/billing active and arrange backup cover. Dang verifies backend/gates/alerts; Duong verifies MySQL/storage/backups; Tai verifies frontend/proxy; Vu signs release acceptance. Owner confirmations and the final protection screenshot remain pending. Release v1.0 was published on 13 November 2026 from green main 0431a5d463295f6acd38b87c367e79c47a7037cc.